

Research and intelligent development of pipeline external welding equipment and sensing technology

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ABSTRACT

The welding efficiency and formation quality of circular butt joints at long-distance oil and gas pipeline installation sites critically influence the construction speed and service performance. Improving the efficiency of welding equipment and enhancing the intelligence of the welding process are crucial for advancing pipeline external welding technology and equipment. In this paper, the current state of pipeline external welding technology and equipment, including the evolution of pipeline welding techniques, implementation of automated welding equipment, and projected future advancements, is reviewed. Welding sensing technologies are discussed, with a focus on the research in vision sensing technology. The application of vision sensing in various welding scenarios, including complex field environments, reveals that its advancement depends on effective sensor design, robust image processing algorithms, and accurate parameter estimation methods. On this basis, the integration of additional sensing technologies is explored to enhance welding perception. Further, the development of an intelligent welding system in which multi-sensor technologies are combined with optimized welding equipment is proposed. This system enables intelligent control of the welding torch position and posture and welding parameters. Leveraging rapidly advancing intelligent technologies, particularly deep learning, can further enhance the intelligence level of the system. Through the strategic design of network architectures and functional modules, along with attention mechanisms and temporal models, the system can effectively learn and focus on critical and historical information from multi-source data sequences. This facilitates the intelligent extraction and precise evaluation of welding feature variables, serving as a key research objective for future pipeline welding and a major advancement direction in the intelligent welding domain.

1. Introduction

Long-distance oil and gas transportation pipelines are essential conduits for energy supply and critical national infrastructure, closely tied to economic growth and public welfare. By the end of 2023, China had constructed approximately 190,000 km of long-distance oil and gas pipelines, including 124,000 km for natural gas (ranking fourth globally), 33,000 km for crude oil, and 33,000 km for refined oil products [1]. Driven by the rising demand for clean energy sources such as natural gas [2] and strong governmental policy support, China's oil and gas pipeline network is projected to continue its rapid expansion in the coming years.

The welding of the circular butt joints is a crucial step in the on-site construction of long-distance oil and gas pipelines. It primarily involves

processes such as pipe end groove machining, fit-up assembly, internal root welding, and external welding (including hot pass welding, filler pass welding, and capping pass welding, as shown in Fig. 1). The external welding efficiency and weld quality of circular butt joints are critical factors that determine the on-site laying speed and the service performance of oil and gas pipelines.

During on-site pipeline installation and circular butt joint welding, the construction environment is highly variable, and welding conditions are often harsh. Because the pipeline maintains a fixed posture after fit-up, arc fusion welding must be performed circumferentially, typically employing downward welding. However, on-site welding scenarios can compromise process stability and weld quality owing to factors such as groove machining deviations, inconsistent assembly gaps, and welding-induced thermal deformation. Consequently, operators need to

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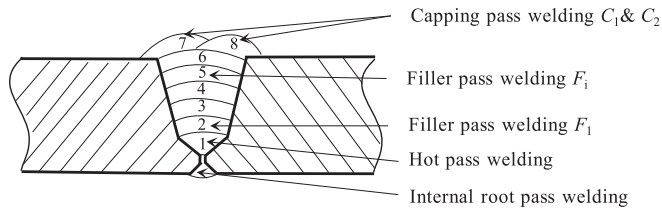


Fig. 1. Weld bead arrangement for double V-shaped groove.

meticulously monitor and adjust equipment and process parameters to ensure that welding behavior and weld quality meet the required standards. Nevertheless, these necessary adjustments still inevitably impact welding efficiency and joint qualification rates.

Nomenclature

WTPP	Welding Torch Position and Posture
DOF	Degree of Freedom
DL	Deep Learning
STT	Surface Tension Transfer
CMT	Cold Metal Transfer
RMD	Regulated Metal Deposition
FCAW	Flux Cored Arc Welding
GMAW	Gas Metal Arc Welding
HAZ	Heat-affected Zone
IMU	Inertial Measurement Unit
RANSAC	Random Sample Consensus
SVM	Support Vector Machine
ROI	Region of Interest
GAN	Generative Adversarial Network
BP	Back Propagation
CNN	Convolutional Neural Network
RNN	Recurrent Neural Network
RBF	Radial Basis Function
ANN	Artificial Neural Network
DCNN	Deep Convolutional Neural Network
LSTM	Long Short-Term Memory
VGG	Visual Geometry Group
YOLO	You Only Look Once
RCNN	Region-based Convolutional Neural Network
FCN	Fully Convolutional Network
TNN	Tensor Neural Network
ERFNet	Efficient Residual Factorized ConvNet
SSD	Single Shot MultiBox Detector
GC	Global Context
ResNet	Residual Network
GRU	Gated Recurrent Unit
CBAM	Convolutional Block Attention Module
MPC	Model Predictive Control

In addition, the all-position downward welding of circular butt joints involves continuous transitions through various welding positions—from flat to vertical and then to overhead—throughout the process. At vertical and overhead positions, the molten pool tends to flow downward under gravity [3], potentially causing process instability and degrading weld quality [4]. The welding torch's tilt angle relative to the welding direction plays a crucial role in determining weld quality. When the torch tilt angle is appropriately adjusted based on the welded position of the circular butt joint, the arc force can partially counterbalance the gravity-induced downward flow of the molten pool, thereby enhancing weld quality during the all-position pipeline downward welding process [5,6]. When laying a pipeline on a slope, such as in hilly terrain, the torch oscillation angle must be correctly set relative to the groove side wall, considering the pipeline's inclination angle and groove size to achieve the desired weld bead shape. Therefore, to satisfy the control requirements of the WTPP during the all-position welding of pipeline circular butt joints with arbitrary inclination angles, the welding robot must possess 5 DOFs, including 3 DOFs for position adjustment and 2 DOFs for posture adjustment.

Achieving full automation and intelligent on-site external welding for pipeline circular butt joints has become a primary research and development objective to improve pipeline laying efficiency and weld quality. The integration of multiple sensing technologies, especially vision sensing, combined with the realization of feature extraction, data fusion, and pattern recognition under complex welding conditions, enables online detection of key parameters during the pipeline all-position welding process. Subsequently, through targeted decision-making strategies, real-time adaptive control of welding equipment, including welding robots and process parameters, can be achieved, reducing operator labor intensity and ensuring high weld quality [7,8]. Further, with the rapid advancement of intelligent technologies such as DL, more accurate and stable feature extraction is possible by leveraging large volumes of sensing data. Meanwhile, attention mechanisms and temporal models are adopted to learn from multi-source data sequences, focusing on key and historical information. These technologies enable precise prediction of welding process parameters and improve the accuracy of intelligent welding task execution. Consequently, intelligent control of the welding process is effectively achieved. These capabilities form a robust foundation for full automation and intelligence in welding, remaining a significant research focus [9–11]. Other emerging technologies such as big data analysis and digital twin systems provide tools to enhance system-level modeling, real-time monitoring, and predictive control in intelligent welding applications.

2. Development and application of on-site pipeline welding technology and equipment

The on-site arc welding process for long-distance oil and gas pipelines has progressed through three primary development stages: manual arc welding, semi-automatic arc welding, and automatic arc welding. As pipeline steel grades improve and pipe diameters and wall thicknesses increase, automatic arc welding has become the dominant method for circular butt joint welding at pipeline laying sites. In developed regions such as Europe and the United States, the welding automation rate for long-distance oil and gas pipeline laying has reached 80 %. However, in China, the welding automation rate is only approximately 40 %. Fig. 2 shows the development of pipeline construction and circular butt joint welding technology in China [12,13].

Continuous advancements in welding technology have resulted in numerous techniques and equipment that enhance welding efficiency and weld quality, which have been extensively studied and applied in the external welding of pipeline circular butt joints. The advancement of automatic welding technology emphasizes integrating online sensing technologies to develop real-time, feedback-controlled welding systems, thereby enabling intelligent welding process control. This section primarily examines the development and application of pipeline external welding technologies and equipment, alongside the progression of intelligent welding systems incorporating online sensing capabilities.

2.1. Arc welding technology

In the early stages of on-site pipeline installation, manual arc welding methods, including electrode arc welding and manual tungsten inert gas arc welding, were prevalent. Currently, their usage has declined in filling and capping welding processes. However, owing to their flexibility, simplicity, strong adaptability, and reliable performance in root welding, they remain employed in constructing small-diameter pipelines and pipeline stations.

Semi-automatic arc welding effectively enhances the efficiency of pipeline external welding and encompasses 4 representative process technologies: STT gas-shielded solid wire arc welding, CMT gas-shielded solid wire arc welding, RMD gas-shielded metal-cored wire arc welding, and self-shielded FCAW. These technologies improve arc welding process stability, increase welding efficiency, enhance environmental adaptability, and ensure high weld quality, and they have been widely

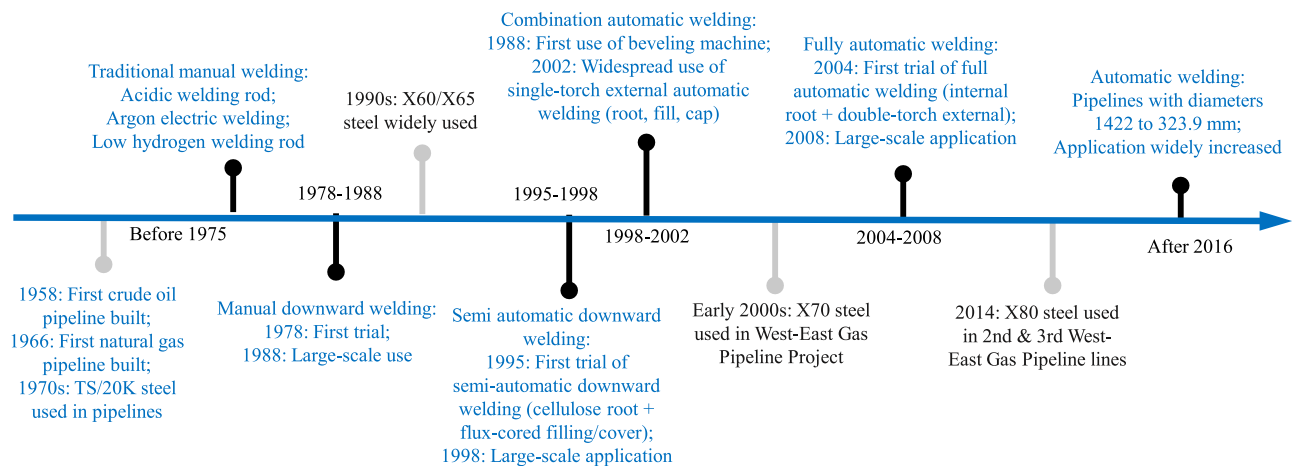


Fig. 2. Development history of pipeline construction and circular butt joint welding technology in China.

applied in long-distance pipeline projects. Applying automated welding equipment at pipeline laying sites can effectively reduce reliance on the technical proficiency of welding operators. The most commonly used automatic pipeline arc welding technologies primarily include FCAW and GMAW. Currently, all-position automatic welding of pipeline circular butt joints is predominantly carried out using orbital welding robots. These robots typically include a multi-DOF actuator, track system, motion controller, welding power supply, wire feeder, and shielding gas system. In response to the increasing demand for welding efficiency, automatic welding systems have evolved from single-torch (single or dual-wire) configurations to dual-torch (single or dual-wire) and multi-torch setups, as shown in Fig. 3 [14].

2.2. Other pipeline welding technologies

In recent years, research institutions worldwide have extensively studied traditional arc welding processes and equipment and other pipeline welding technologies to improve welding efficiency, accelerate pipeline installation, and reduce HAZ softening. These technologies include submerged arc welding, high-energy beam welding (e.g., laser welding, electron beam welding, and laser-arc hybrid welding), friction welding (e.g., radial friction welding and friction stir welding), and flash butt welding. The characteristics and limitations of these welding techniques are detailed in Table 1.

Table 1 Other welding technologies for pipeline applications.		
Welding technology	Application characteristics	Application limitation
Submerged arc welding	Commonly used in welding stations for the welding of double or triple-connected pipes	Difficulty in ensuring joint quality and transporting multiple pipes
Laser welding	High energy density and a small HAZ significantly improve pipeline welding efficiency	Requires high precision in pipeline fit-up assembly owing to the small laser focal point
Electron beam welding	Fast welding and high weld quality for pipelines with thick walls	Generates excessive X-ray radiation during operation
Laser-arc hybrid welding	1. Combining the advantages of laser welding and arc welding; 2. Minimal workpiece deformation and low precision requirements for pipeline fit-up	Challenges in energy supply for on-site applications, and the laser equipment size is a constraint
Radial friction welding	Stable welding process	Limited to specific pipeline diameters
Friction stir welding	Produces welded joints with excellent mechanical properties	Challenges in the design and control of the stirring head
Flash butt welding	High automation, one-time molding, high production efficiency, and considerable potential for large-diameter and thick-walled pipelines	Challenges in ensuring weld quality; currently at the experimental stage

2.3. Pipeline external welding equipment and its intelligent development

2.3.1. Evolution of automated pipeline external welding equipment

As early as the late 1960s, single-torch all-position automatic welding equipment utilizing GMAW was developed, as shown in Fig. 3a. It was first successfully applied in pipeline construction and was subsequently adopted and refined by institutions in various regions. To further improve welding efficiency, dual-wire arc welding technology

was introduced, arranging two wires in tandem within a single-welding torch, resulting in approximately 15 % greater efficiency compared with single-wire systems. Since the 1980s, dual-torch automatic pipeline welding systems have been developed, as shown in Fig. 3b, combining the advantages of multi-wire welding and improved arc utilization to enhance overall productivity and cost-effectiveness in pipeline welding.

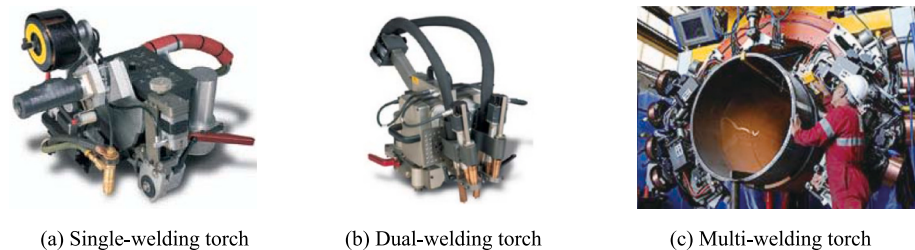


Fig. 3. Classification of all-position automatic welding equipment [14].

Additionally, automated multi-torch welding equipment was designed for large-diameter pipelines (≥ 813 mm external diameter) to increase welding speed. Fig. 3c shows the 8-torch automatic external welding equipment, which can simultaneously operate 4 welding heads (each with dual-torch), significantly improving the welding efficiency.

As the number of welding torches and wires increased, the external welding efficiency of pipelines increased markedly. During this process, the heat input at each welding position can be effectively controlled by appropriately adjusting the spacing and current ratio between tandem wires, employing interleaved and segmented multi-torch welding techniques, and optimizing welding parameters (including welding speed, arc voltage, and welding current). These measures effectively mitigate HAZ softening and the associated decreases in hardness and toughness [15–17].

Since the 1970s, China has been researching and developing pipeline automatic welding equipment and supporting processes, progressively advancing from single-torch to dual-torch automatic welding systems. These systems have markedly improved construction efficiency and weld quality and have been widely adopted in major pipeline projects both domestically and internationally in recent years.

Automated pipeline welding systems have significantly evolved over the past few decades, with global technological solutions addressing challenges related to productivity, process stability, and weld quality. Table 2 presents a comparative analysis of various automated pipeline welding systems.

2.3.2. Structural design and actuation mechanisms of pipeline external welding robots

Regarding the mechanical structure of pipeline welding robots, Luo et al. [18] and Zhou et al. [19] developed walking mechanism and welding torch adjustment mechanisms for orbital robots. These robots feature 3 DOFs: walking, welding torch lateral adjustment, and vertical adjustment, as shown in Fig. 4a. They possess essential capabilities for pipeline welding, including weld seam tracking through lateral position adjustment and maintaining arc stability by regulating vertical torch height during the welding process. To meet the posture DOF requirements for all-position pipeline welding, Li et al. [20] added a tilt angle adjustment to the original 3 DOFs, creating a 4-DOF welding robot mechanism. This additional DOF enables the generation of an optimal molten pool force field by controlling the torch tilt angle. Cao et al. [21] incorporated a welding torch oscillation mechanism into the pipeline welding robot, as shown in Fig. 4b, allowing for welding torch angle swinging and edge stopping during welding. However, neither design fully satisfies the comprehensive requirements for WTPP control, indicating their inability to fulfill the complete adjustment needs for WTPP during all-position pipeline welding with arbitrary inclination angles. To address the need for WTPP control, Zhu et al. [22] designed a 5-DOF pipeline welding robot based on the Pieper criterion. This robot

includes 3 position DOFs: robot walking DOF D_v , welding torch lateral adjustment DOF D_e , and vertical adjustment DOF D_h , along with 2 posture DOFs: torch tilt angle adjustment DOF D_α and oscillation angle adjustment DOF D_β , as shown in Fig. 4c. The designed mechanisms comply with the Pieper criterion, wherein the two rotational joint axes controlling the torch tilt and oscillation angles converge at a single point, through which the welding torch axis also passes. This arrangement facilitates a closed-form solution to the inverse kinematics problem, thereby preventing multiple solutions, accelerating computation, and simplifying joint motion control. Consequently, it achieves comprehensive WTPP control, fulfilling the requirements for welding process stability and weld formation quality. The comparison between the DOF of various robotic mechanisms in Table 3 clearly reveals their WTPP adjustment capabilities. Ongoing advancements in robotic actuation mechanism design have established a robust foundation for the flexible control of WTPP, offering enhanced adaptability in complex and variable-posture welding scenarios.

2.3.3. Intelligent sensing and control in on-site pipeline welding robots

Realizing intelligent pipeline welding depends on integrating online sensing. Because online sensing technology is combined with automatic welding systems, welding robots can execute real-time feedback control based on sensor data during the welding process, enabling intelligent tasks such as weld seam tracking, welding groove parameter detection, welding torch positioning, and relative posture control [23–27].

Pukyong University in South Korea developed a welding robot equipped with a mechanical contact-type seam tracking sensor. Integrating a distance sensor enables automatic welding of planar grid-type steel structures [28]. Keio University in Japan developed a mobile welding robot equipped with a vision sensor. It can move along the pipeline, detect the weld seam through vision sensing, and control the welding torch to achieve automatic weld seam tracking [29]. Companies from various countries have also developed structured light vision sensors, offering technical and product support for vision-based welding applications.

Nanchang University designed a trackless welding robot that magnetically adheres to steel structures and employs a laser vision sensor for automatic weld seam tracking during wall climbing and welding [30]. The Beijing Institute of Petrochemical Technology developed a permanent magnet-wheeled robot featuring a line laser sensor for autonomous welding, utilized in water supply pipeline construction and large spherical tank welding [31]. Shanghai Jiao Tong University achieved autonomous weld seam tracking and welding of medium-thick plates using laser vision sensing [32,33], as shown in Fig. 5a. They also utilized passive vision sensing to track weld seams by extracting the weld contour center [34,35]. The South China University of Technology systematically explored welding feature point extraction and seam tracking under strong arc light interference using a three-line laser vision sensor (Fig. 5b), providing a solution for arc welding applications [36–39]. Tsinghua University developed a trackless crawling welding robot with a hybrid wheel-track mechanism. The robot is equipped with a laser vision sensor and has been successfully utilized for automatic weld seam tracking on large spherical tanks and ship hulls [40]. In practical applications, trackless welding robots must feature a lightweight structural design and an optimized mass distribution center to reduce traction loss and prevent inadequate magnetic adhesion, thereby enhancing overall system stability.

Regarding the intelligence of pipeline welding robots, Zhu et al. [22,41] developed an intelligent pipeline welding system that incorporates a dual-line structured light sensor, a dual-axis tilt sensor, and a 5-DOF welding robot, as shown in Fig. 5c. This system can detect welding groove sizes, WTPP parameters, and pipeline posture in real time. Consequently, during all-position pipeline welding, it enables real-time correction of the torch lateral deviation e to ensure precise weld seam tracking, adjustment of torch height H to maintain arc stability, planning of the torch tilt angle α based on pipeline position to generate

Table 2
Comparative analysis of automated pipeline welding systems.

Developer	Welding method	Torch configuration	Key features
US	GMAW	Single-torch	First field application; early automation
Austria, France, Italy	GMAW	Single-torch, early Tandem	Improved control systems; regional applications
Germany	Tandem GMAW	Single-torch, dual-wire	~15 % efficiency improvement; front-rear wire synergy
UK	Dual-torch GMAW	Dual-torch, dual-wire	Enhanced productivity and heat input control
France	Multi-torch GMAW	4 heads $\times$ dual-torch	High productivity; suitable for ≥ 813 mm pipelines
China	GMAW	Single-torch, dual-torch	Widely applied in major pipeline projects

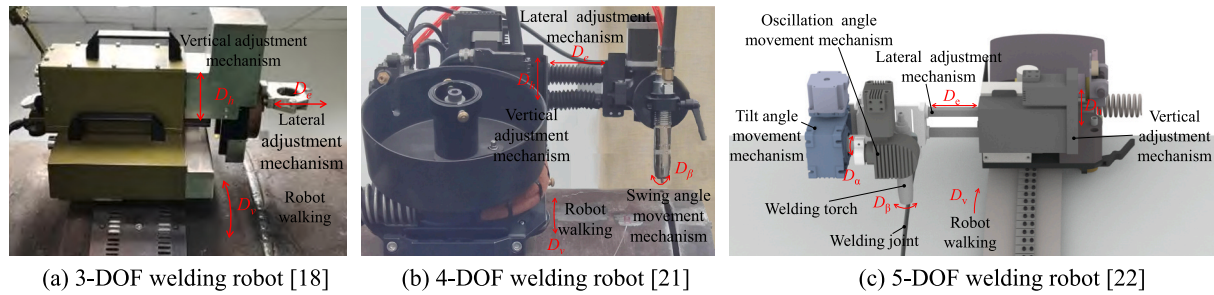


Fig. 4. Pipeline welding robots with different degrees of freedom [18,21,22].

Table 3

Comparison between different robotic mechanisms' degrees of freedom.

Robotic system Mechanism name	The robot in [18,19]	The robot in [20]	The robot in [21]	The robot in [22]
Walking mechanism (D_w)	✓	✓	✓	✓
Vertical adjustment mechanism (D_v)	✓	✓	✓	✓
Lateral adjustment mechanism (D_h)	✓	✓	✓	✓
Tilt angle adjustment mechanism (D_t)	×	✓	×	✓
Oscillation angle adjustment mechanism (D_o)	×	×	✓	✓

optimal arc force and improve weld formation quality, and setting of the torch oscillation angle β according to groove size to ensure effective weld bead fusion. A schematic of the WTPP adjustments during all-position pipeline welding is presented in Fig. 6. Correcting the torch position and adjusting the torch's posture angle can effectively reduce defects such as insufficient sidewall fusion, thereby increasing the weld formation qualification rate. Concurrently, adjusting the torch tilt angle can mitigate issues such as excessive reinforcement caused by molten pool flow at critical welded positions in pipelines, thereby ensuring consistent weld formation [42]. In summary, all-position pipeline welding utilizing coordinated 5-DOF torch control can significantly enhance welding efficiency and improve weld quality.

When developing the intelligent external welding system, the accuracy and stability of torch motion control must be prioritized. Multi-sensor fusion navigation (including vision, IMU, and laser) enhances the perception and tracking of weld paths on curved surfaces, thereby improving path-following accuracy. In addition, DL techniques are employed for curved path prediction and adaptive trajectory planning, enabling the system to anticipate and respond to path variations more robustly. For motion control, a hybrid approach that combines feed-forward trajectory prediction with real-time feedback correction ensures

both the accuracy and stability of torch movement. Furthermore, DL techniques extract and integrate data from various sensors, allowing intelligent welding systems to perform real-time monitoring and control of molten pool temperature, welding current, arc voltage, and WTPP. This facilitates dynamic adjustment of process parameters, thereby ensuring high-quality and consistent weld formation.

In summary, other researched pipeline welding technologies have not been widely adopted owing to their limitations. Arc welding remains the most commonly used technology for external welding of circular butt joints in pipeline field construction. To enhance the efficiency, automation, and intelligence of pipeline butt joint welding, future development directions encompass: 1. adopting advanced welding technologies, such as laser-arc hybrid welding, and addressing on-site application challenges to adapt them for pipeline external welding; 2. refining equipment and arc welding methods to boost pipeline welding efficiency, for instance, employing multi-torch (or multi-wire) welding technology with controlled heat input; 3. integrating multi-sensing technologies for real-time information detection and combining them with DL techniques for multi-parameter joint computation to advance intelligent pipeline external welding. These initiatives will promote the development of an integrated intelligent pipeline welding system, ensuring real-time adjustment of WTPP and welding process parameters during pipeline welding. Consequently, intelligent welding of pipeline circular butt joints can be realized.

3. Sensing technologies and applications in welding

3.1. Classification of sensing technology in welding

The welding process can be monitored with sensors, and real-time control can be implemented based on online sensing feedback. With advancements in computer technology and the ongoing development of various sensing technologies, leveraging sensing information to achieve intelligent control of the welding process has become a key trend in the welding domain. During the arc welding process, numerous sources of interference exist, such as metal spatter, intense arc light, smoke, and dust, all of which necessitate sensors with robust anti-interference

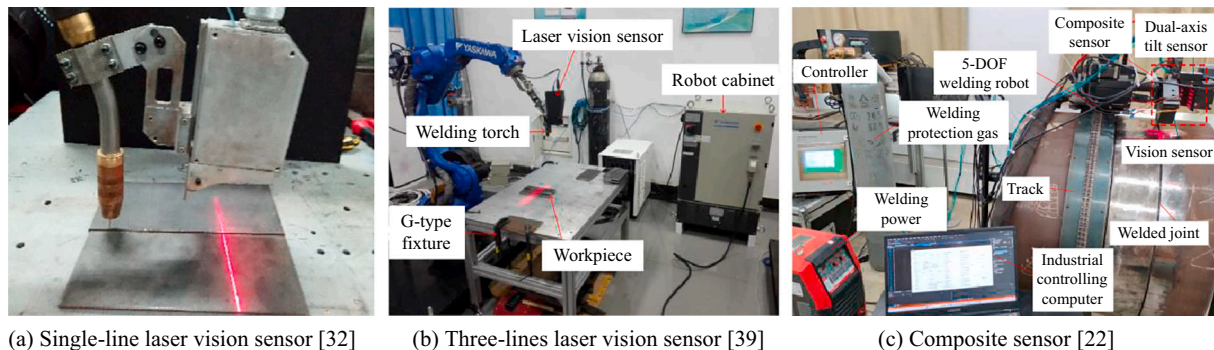


Fig. 5. Welding robot based on online sensing [22,32,39].

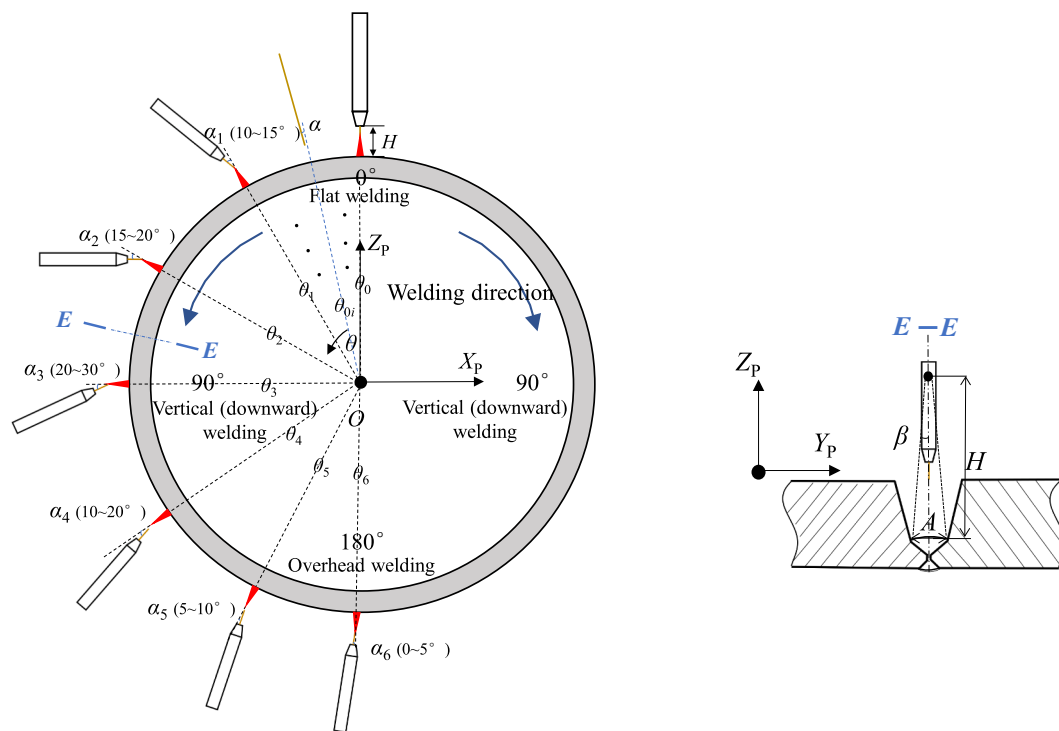


Fig. 6. Schematic of WTPP adjustments for all-position pipeline welding.

capabilities. Currently, the applicable sensing technologies in the arc welding domain primarily include contact sensing, arc sensing, vision sensing, ultrasonic sensing, and tilt angle sensing, as shown in Fig. 7 [43–45]. These technologies are primarily utilized for weld seam tracking, welding groove size detection, and WTPP parameter calculation.

Most of the information welders rely on to monitor and control the arc welding process is derived from visual perception. Compared with other sensing technologies, vision-based sensing offers non-contact operation, rich detection data, high accuracy, and sensitivity. Vision sensing has been extensively researched and applied across various domains, including weld seam tracking, welding groove parameter detection, welding torch positioning, and relative posture detection [25,26,46–48]. Additionally, integrating vision sensing with other sensing methods continues to expand its applications and enhance the performance of intelligent welding systems.

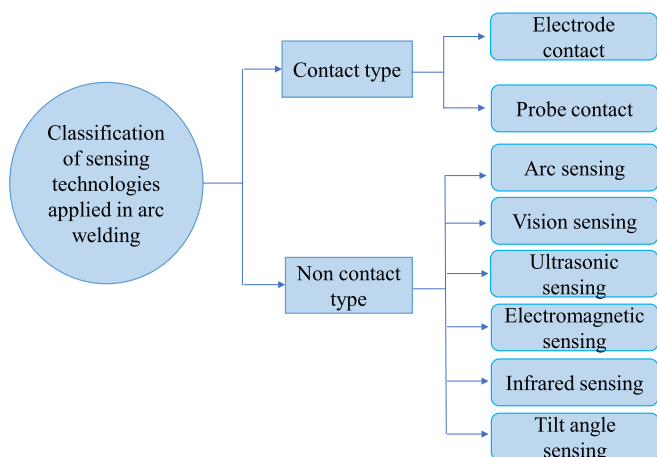


Fig. 7. Classification of sensing technologies applied in arc welding.

3.2. Vision sensing technology and its applications

3.2.1. Classification of vision sensing technology

Vision sensing technology is classified into two categories: active vision and passive vision, based on the utilization of external lighting sources. Passive vision relies on ambient or arc light for background illumination, capturing images of the welded area through industrial cameras. In contrast, active vision employs external auxiliary light sources projected onto the welded region and adjacent surfaces of the workpiece, enabling industrial cameras to record the modulated feature information of the welded area. Owing to the advantages of strong directionality, monochromaticity, high coherence, and energy concentration, laser-structured light is commonly used as an auxiliary light source. The schematics of passive and active vision are shown in Fig. 8.

Applications of vision sensing technology in welding are categorized into three scenarios: “Pre-weld,” “In-weld,” and “Post-weld,” as presented in Table 4. During the “Pre-weld” phase, vision sensing is primarily utilized for initial welding point recognition [49–52], welding groove type identification [53,54], and welding torch positioning [26,55], thereby guiding the motion control of the welding torch before the welding process begins. During the “In-weld” phase, it is mainly employed to measure groove size parameters [56,57], identify WTPP, achieve weld seam tracking [34,58–67], monitor arc behavior, and

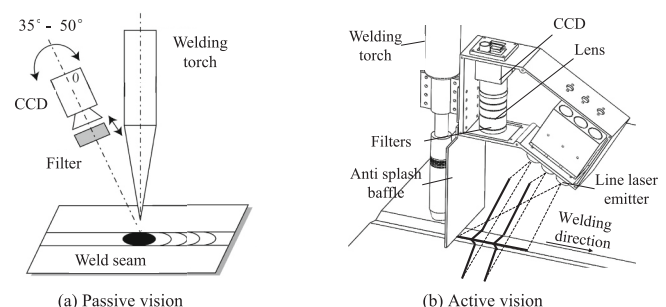


Fig. 8. Schematic of vision sensing system.

Table 4
Classification of vision-sensing applications in the welding domain.

Application scenarios	Specific applications	Method/algorithm adopted	Sensing type
Pre-weld	Starting position recognition	Boundary curve fitting and corner detection [49]	Passive vision
		Edge detection and corner coordinate extraction [50]	
		Template matching and line fitting [51]	
	Groove type identification	Center contour extraction and RANSAC normal extraction [52]	Line structured light vision
		Laser feature point coordinate values calculation [53]	
		Weld seam type recognition based on SVM [54]	
In-weld	Welding torch positioning and posture recognition	Cross-line laser sensing [55]	Line structured light vision
		Surface reconstruction for flat workpieces [26]	
		Groove feature point calculation based on multi-line structured light [56,57]	
	Welding size detection	Surface reconstruction for planar workpieces based on dual-line structured light [26]	Line structured light vision
		Feature extraction and linear fitting [58–61]	
		Target tracking based on Gaussian kernel [62]	
	Weld seam tracking	Continuous convolutional operator target tracking [63]	Active vision
		Cross laser line detection and line fitting [64]	
		Circular laser line detection and feature extraction [65]	
		Feature extraction and edge detection [34,66]	
Post-weld	Weld pool feature detection	Statistical filtering and line fitting [67]	Passive vision
		Contour extraction based on Bayesian classifier [68]	
		Back molten pool contour extraction based on directional light [69]	
	Surface quality and defect detection	Molten pool depression extraction based on line structured light [70]	Active vision
		3D contour reconstruction of the weld bead and mathematical model establishment [71,72]	
		3D reconstruction of weld bead and point cloud segmentation [73]	

detect molten pool characteristics [68–70], thereby facilitating the adjustment of welding parameters to ensure high weld quality. During the “Post-weld” phase, the process primarily inspects the surface formation quality and potential external defects of the weld bead [71–73], assesses the weld seam qualification rate, and guides the repair of defective welds.

3.2.2. Structured light vision sensing technology and its applications

The active vision based on monocular structured light has advantages of straightforward image processing, high detection accuracy, and robust anti-interference capabilities. Consequently, it is widely used in weld seam tracking [34,62–67], groove size detection [56,57], and

other applications [59]. These systems are classified based on measurement principles, structured light type, structural design, installation position, application scenario, and purpose, as shown in Fig. 9 [43–45,51–57,68–70,74].

The central challenge in employing structured light vision sensing for detection lies in identifying the deformation characteristics of the laser-structured light projected onto the welding groove. A primary application is welding groove type recognition, which facilitates groove size detection and weld seam tracking. Qian et al. [53] proposed an automatic weld type recognition method using structured light vision sensing by determining the feature point coordinates of the deformed laser stripe. However, its complexity and high computational load compromise real-time performance. To enhance efficiency, Fan et al. [54] developed an SVM-based system for automatic weld seam type identification, utilizing laser stripe features from six groove types as input vectors for classification. For welding groove size detection, Guo et al. [56,57] designed an anti-interference detection method employing a structured light vision sensor for V-groove size measurement. Zhu et al. [22,26] utilized a dual-line structured light sensor to reconstruct groove surfaces through data segmentation, point cloud mapping, and plane fitting (illustrated in Fig. 10). This approach achieved a high precision of 0.1 mm in measuring groove dimensions and determining the welding torch relative position and posture parameters. Structured light vision sensors are also extensively employed for weld seam tracking. Zhao et al. [75] developed a system using single-line structured light sensing to monitor lap, fillet, and butt joints, attaining a tracking accuracy of up to 0.69 mm and a misalignment detection accuracy of 0.2 mm. In [53–55,58,59], line structured light sensors were employed to extract deformed laser features, perform linear fitting, and identify the seam center point to control the welding torch for weld seam tracking. Additionally, cross and circular laser sensors provide comprehensive geometric and positional data, which also have been applied in weld seam tracking [64,65].

During the welding of spatial posture welded joints, the relative posture of the welding torch critically influences weld quality and must be adjusted accordingly. Mao et al. [55] used a cross dual-line laser-structured light sensor to detect the welding torch’s height, weld position, and trajectory direction, enabling the welding robot to execute automatic welding in the optimal posture. Zhu et al. [26,76] designed a dual-line structured light vision sensor to ascertain the position and posture of the welding torch relative to the workpiece, providing a basis for real-time WTPP control.

After welding is completed, line structured light vision sensors assess weld surface morphology and defects to determine weld formation quality and guide repairs. Li et al. [71] introduced a laser-structured light sensing technique to capture 3D weld bead contours, extracting key parameters such as weld width, filling depth, and groove width to evaluate weld bead quality. Chu et al. [72] developed a post-welding inspection method using structured light sensors to generate a weld seam point cloud. They established a mathematical model to detect weld width, reinforcement height, and misalignment, effectively enabling online weld defect detection.

The application of structured light vision sensing technology in welding has been extensively researched, enhancing the automation of welding equipment and facilitating intelligent welding processes.

3.2.3. Research on image processing technology in the welding domain

In the welding domain, whether employing passive or active vision sensing, the critical factor for detecting weld groove (weld seam) features and intelligently controlling the welding torch lies in extracting essential feature information from captured images. Because of image ambiguity across various scenarios, developing scene-specific image processing algorithms is essential. These algorithms must ensure robust generalization, high reliability, accuracy, and real-time performance in similar scenarios.

For passive vision sensing detection, image processing algorithms

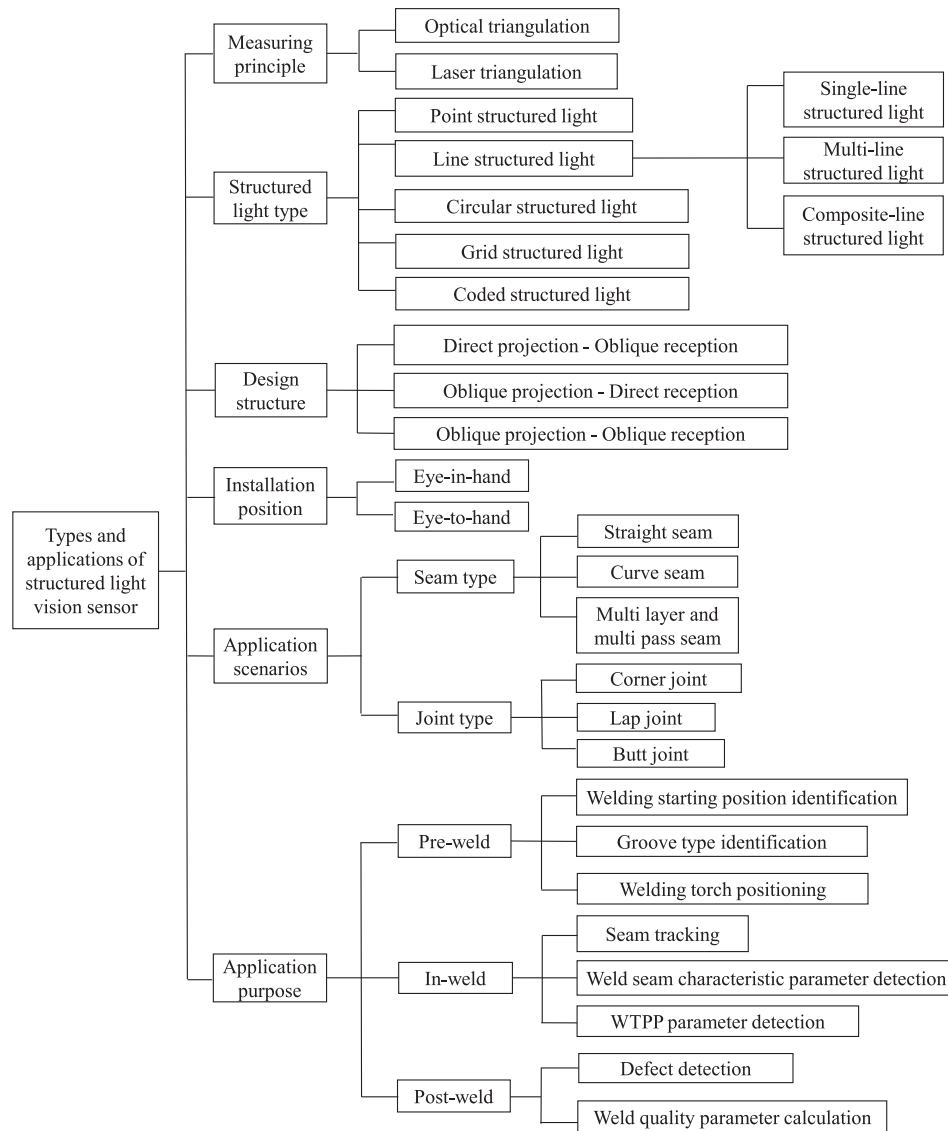


Fig. 9. Types and applications of structured light vision sensor in welding.

vary according to application settings. Chen et al. [49] proposed a two-step coarse-to-fine approach to accurately detect the welding start position. First, they fitted the weld seam and workpiece boundary curves to obtain a preliminary starting point. Subsequently, a small window centered on this point was employed to accurately determine the welding start location via corner detection. Wei et al. [50] developed a morphological corrosion edge detection algorithm to extract weld seam image edges. This algorithm effectively filtered out noise and preserved essential features, ultimately identifying the weld starting point.

For weld seam tracking, numerous researchers have explored various image-processing algorithm designs. Ye et al. [66] employed a double-window approach to separately extract the molten pool center and the weld seam, as shown in Fig. 11. They applied Sobel edge detection for weld seam extraction, achieving satisfactory accuracy in tracking both straight and curved welds. However, improvements in adaptability and real-time performance under complex conditions are needed. Xu et al. [34] assessed the deviation between the weld seam centerline and the welding wire in GMAW by employing edge detection, pseudo-edge removal, and feature extraction from molten pool images. Their experiments demonstrated that the measurement accuracy could reach 0.3 mm.

Line structured light vision sensors are the most extensively used in

active vision sensing, with their image processing objectives being relatively consistent—aiming to extract point or line features from deformed laser line images. Numerous researchers have investigated image extraction algorithms tailored to the deformed laser line imaging characteristics when line structured light is projected onto a single V-groove. In [58–61], an image processing workflow was designed for single V-grooves and included image preprocessing, ROI selection, thinning, linear fitting, and feature point extraction. This workflow ensures precise extraction of deformed laser lines and real-time weld seam tracking, as shown in Fig. 12. Zhang et al. [77] introduced a four-step method to accurately identify laser line feature points in Y-shaped grooves, enabling real-time tracking. Zhu et al. [41] presented a partition time–frequency–space processing and fusion method for laser image in double V-grooves, facilitating the rapid and accurate extraction of non-uniform line features. For extracting deformed laser line feature points, Zou et al. [62] improved robustness by using three-line laser-structured light and a Gaussian kernel-based tracking method. They further enhanced tracking accuracy in noisy environments by applying continuous convolution operators, reducing errors to below 0.2 mm [63]. In specific algorithms, multi-line structured light performs better than conventional single-line structured light.

The laser line appears as a curve when the structured light is

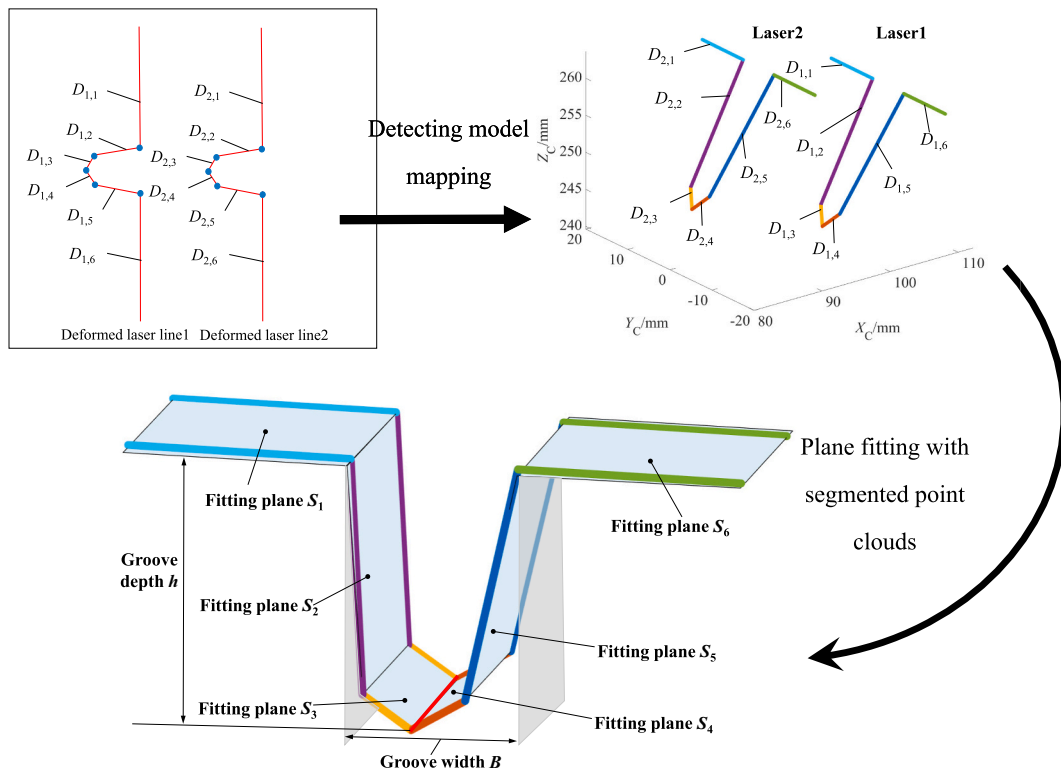


Fig. 10. Reconstruction of welding groove surface based on plane fitting [22].

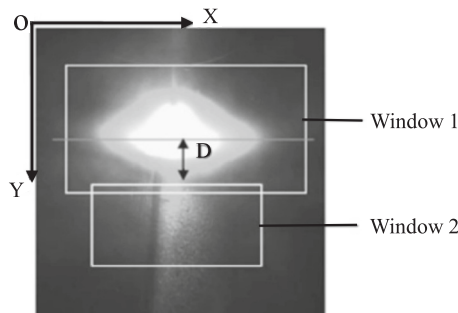


Fig. 11. Double-window segmentation of welding images [66].

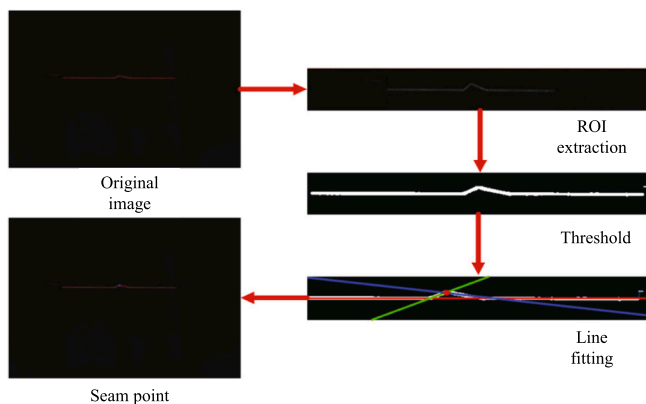


Fig. 12. Main steps of the traditional laser line extraction algorithm [59].

projected onto the weld bead. Peng et al. [70] developed an image-processing program, shown in Fig. 13, to extract molten pool concavity in real time for controlling backside weld bead width. Chu et al. [72] created a mathematical model linking deformed laser lines to the weld bead, allowing the calculation of weld width, reinforcement height, and misalignment, thereby enabling the online detection of weld defects. When employing structured light vision sensors in post-weld inspection systems, the detection accuracy for sub-millimeter surface defects (such as underfill or excessive reinforcement) can be markedly improved through methods including point cloud mesh reconstruction, local curvature and height deviation analysis, sliding-window profile scanning, and DL-based defect recognition on point clouds [78,79]. These enhancements enable more precise identification of subtle surface anomalies, thereby increasing the overall accuracy of defect detection and enhancing the system's robustness and reliability in complex field environments.

Although vision sensing is widely used in welding, its implementation faces numerous interferences and challenges. Achieving stable feature detection under complex conditions remains a primary research objective.

3.2.4. Application analysis of vision sensing in multi-interference environments

In on-site pipeline welding settings, vision sensing frequently contends with issues such as lighting inconsistencies, high reflectivity, arc shadows, wind, dust, and electromagnetic interference. These factors significantly compromise the accuracy, reliability, and stability of sensor detection. Fig. 14 shows the sensor detection process within a complex pipeline welding environment. Researchers have thoroughly explored solutions to attain accurate and stable vision-based feature detection amid multiple interferences.

During on-site welding, particularly in all-position welding, lighting inconsistencies are the primary factor affecting vision sensing, directly influencing feature detection accuracy and stability. Structured light vision sensing typically utilizes laser emitters with specific central

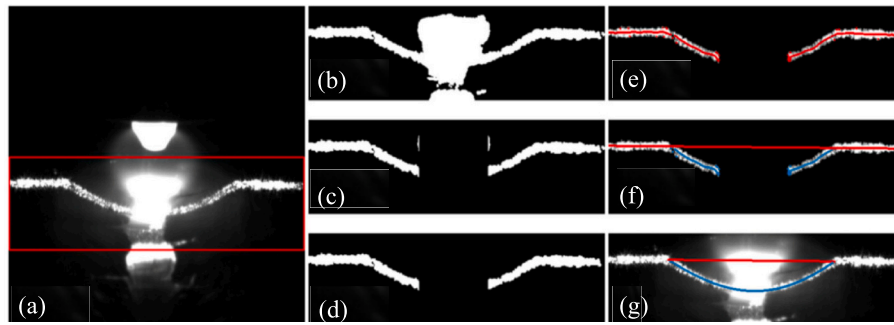


Fig. 13. Image processing: (a) ROI selection; (b) binarization; (c) interference elimination; (d) further eliminate interference; (e) centerline extraction; (f) line detection and pixel classification; (g) curve fitting [70].

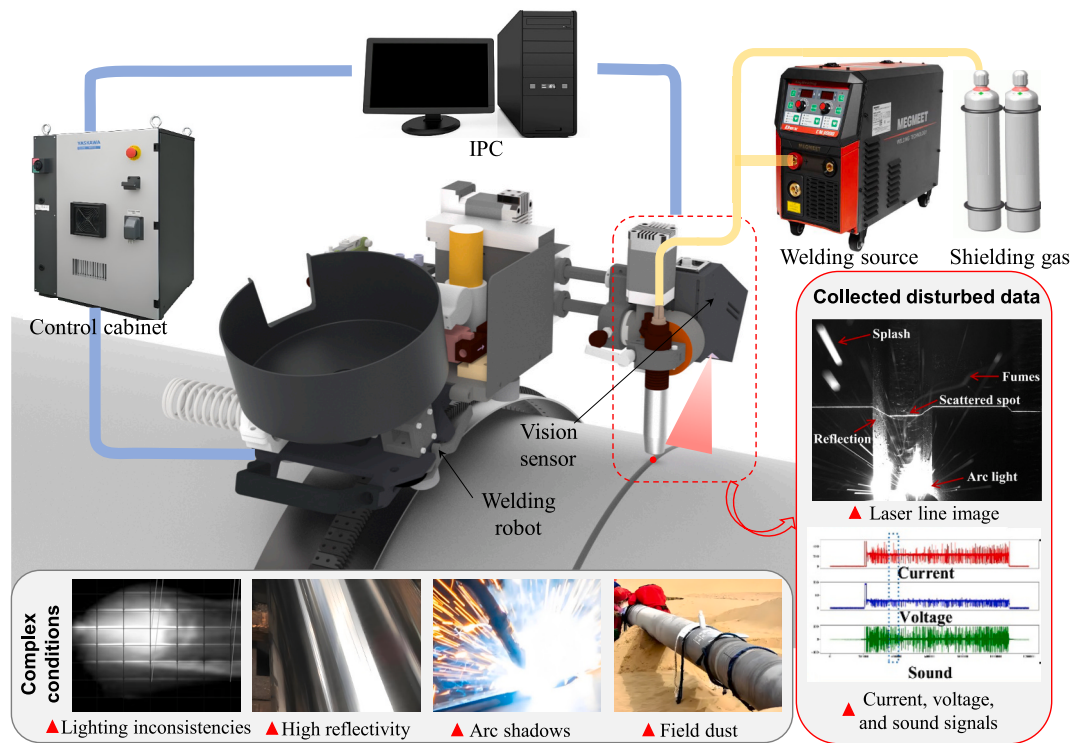


Fig. 14. Sensor signal acquisition under complex pipeline welding conditions.

wavelengths (such as 650 nm, where arc light intensity is comparatively low). When applying structured light vision for measurements under complex lighting conditions, Wang et al. [80] developed a method that integrates difference imaging, region-growing speckle noise removal, and sub-pixel center extraction. This technique effectively removes speckle noise caused by lighting variations and achieves center detection accuracy below 0.1 pixels. The system can capture high-contrast laser stripe images even under uneven lighting conditions by employing adaptive exposure control, image enhancement techniques, and saliency detection, thereby ensuring robust feature extraction accuracy [81,82]. In passive vision systems with inconsistent lighting, researchers typically use image enhancement techniques alongside more robust edge detection algorithms. Examples include Canny edge detection with adaptive thresholds and Hessian-based methods, which enhance curved structures and accurately extract weld seam edges in high-contrast settings [83,84]. Angle variations in all-position welding and the high reflectivity of the workpiece surface near the welding groove can cause strong specular reflectance at certain points. To address this, attaching optical (polarizing) filters to vision sensors effectively reduces specular highlights. Hu et al. [85] applied bilateral and optical filtering to laser

stripe images of reflective base materials in welding environments, suppressing specular highlights while maintaining laser stripe edges and significantly enhancing seam detection robustness under high reflection conditions. Moreover, using multi-angle lighting to prevent strong reflections from a single direction proves effective [86]. In addition, adopting multi-channel image fusion techniques can mitigate interference from highly reflective regions [87]. These strategies have demonstrated effectiveness in weld seam tracking applications.

In arc welding, arc shadowing poses a significant challenge, often leading to substantial image contrast reduction and the occlusion of weld features. Vision-based sensing systems for short-circuit transfer arc welding typically employ arc-synchronized exposure control, dynamically adjusting camera exposure during the short-circuit phase to prevent peak arc illumination [68]. Designing dedicated denoising encoders effectively suppresses image noise from arc light and metal spatter, facilitating clearer extraction of laser stripes. Qin et al. [88] introduced a paired response noise model from the imaging perspective, establishing an inverse response law and corresponding loss function to achieve effective image denoising. Visual attention mechanisms are commonly used for feature detection in these scenarios. These

mechanisms create attention maps to adaptively suppress arc-disturbed regions, thereby mitigating the impact of arc interference on feature extraction [89].

In addition, in complex environments with wind, dust, electromagnetic interference, and extreme temperatures, appropriate physical protective devices and modules can be designed to ensure sensor stability and accuracy. For instance, to counter wind-induced vibration, water, and dust, wind-resistant structures enhance mechanical stability, while waterproof and dustproof housings and protective coatings provide environmental sealing. To handle extreme temperatures, industrial-grade components are utilized alongside internal thermal control systems and insulation layers. Furthermore, metal shielding, proper grounding, fiber optic transmission, and differential signal processing can improve system stability and robustness in high electromagnetic interference environments. These integrated strategies effectively sustain sensor performance under adverse conditions.

Moreover, by integrating data from multiple sensors and applying DL-driven data fusion techniques—such as attention mechanisms, GANs, and invariant feature learning—the system can effectively navigate challenges in complex environments, including pose variation, reflection, occlusion, and lighting interference [90,91]. In particular, attention-based models can dynamically concentrate on undisturbed image regions, suppressing areas plagued by strong interference and thereby improving feature localization accuracy. GANs are also utilized to reconstruct or enhance distorted weld images, effectively restoring critical structural features lost to intense reflections or occlusion. This furnishes robust technical support for realizing intelligent welding across diverse object conditions (e.g., workpiece type, groove type) and constitutes a promising area for further research.

3.3. Fusion applications of multi-sensor technologies

To further elevate the intelligence of the welding process, additional information detection is essential. Integrating vision sensing with other sensing technologies facilitates more intelligent detection and control of the welding process, advancing the development of intelligent welding systems.

Achieving intelligent robotic welding necessitates real-time control of both the welding torch's position and spatial posture. The adjustment of the torch's forward/backward tilt and left/right oscillation angles depends on the absolute posture of the workpiece. In all-position pipeline welding, the continuously changing welding position requires real-time adjustments of the torch tilt angle to ensure high-quality weld formation. Additionally, the torch oscillation angle must be adjusted according to pipeline inclination and welding groove sizes. Integrating gravity and vision sensing can achieve intelligent WTPP detection, thereby enabling comprehensive control of WTPP for high-quality weld formation. Zhang et al. [76] developed a composite sensor that combines vision and gravity sensing to detect workpiece posture and regulate torch posture during planar welding. Zhu et al. [22] further applied the composite sensor to the all-position pipeline welding process. By detecting welding position and posture parameters alongside a welding torch control algorithm, they achieved real-time WTPP control that meets all-position welding requirements. The composition of the composite sensor is shown in Fig. 15.

Additionally, integrating vision sensing with other technologies facilitates comprehensive information perception during welding, thus providing technical support for high weld quality. Zuo [92] merged visual data with arc signals to explore the relationships among GMAW electrical signals, molten pool characteristics, and weld defects, enabling intelligent defect recognition. Chen et al. [93] used arc and vision sensors to collect pulsed GTAW welding data and applied a BP neural network for multi-source data fusion, which improved the accuracy of predicting backside weld width compared with single sensor models. Lei et al. [94] introduced a multi-source sensor consisting of a monocular camera vision sensor, a laser rangefinder, and an arc sensor.

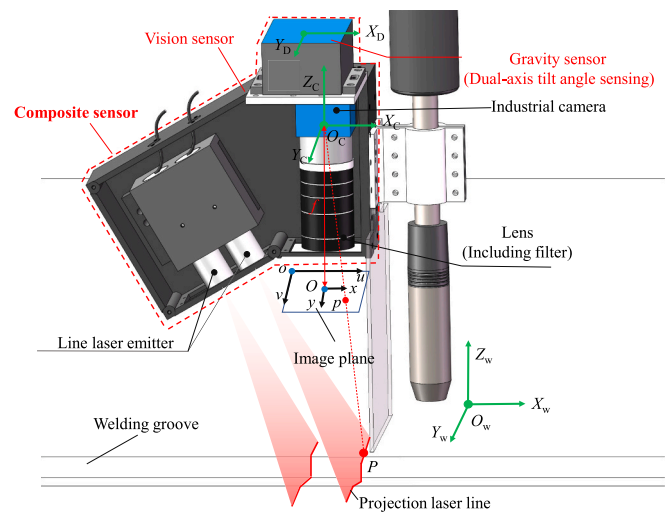


Fig. 15. Composite sensor based on visual sensing and gravity sensing [22,76].

The vision sensor identifies the weld seam center, the laser rangefinder determines the welding torch height, and the arc sensor adjusts the arc length accordingly. Through the generation of a tube sheet point cloud model, precise weld seam detection was achieved. To predict welding states—such as proper fusion, lack of fusion, and burn-through—in T-joint double-sided double-arc welding, Xu et al. [91] proposed an online monitoring method that fuses welding current, arc voltage, arc sound, and molten pool images. A CNN-based molten pool feature extractor was developed to support the system, achieving a prediction accuracy of 98.538 %. The multi-source signal fusion process is illustrated in Fig. 16.

Although multi-sensor fusion for intelligent welding has advanced significantly, several challenges require further investigation. A primary difficulty is the temporal synchronization and spatial alignment of heterogeneous sensor data, especially when integrating asynchronous signals such as vision frames, arc signals, and inertial data. Effective time-series alignment and data preprocessing are essential to ensure meaningful fusion [95]. Additionally, the high dimensionality and redundancy of multi-source data increase computational complexity and may hinder real-time performance. Techniques such as principal component analysis, attention-based feature weighting, or graph neural networks for feature selection or dimensionality reduction can preserve essential features while reducing data burden. Moreover, robust fusion strategies must consider sensor reliability and environmental noise. Research into dynamic weighting methods and confidence modeling can be developed to handle uncertain or degraded sensor inputs [96]. Ultimately, the explainability and interpretability of fusion-based decisions are crucial for industrial implementation, as they facilitate troubleshooting and ensure regulatory compliance in safety-critical welding applications [97].

Future research can incorporate additional rich data, such as temperature measurements and molten pool oscillation signals, into the fusion process to further refine welding process control and attain superior weld quality. When utilizing artificial intelligence techniques to develop multi-source information fusion models, priority should be given to explainable deep fusion architectures and deployment strategies. These methods will enable welding systems to make real-time, robust, and transparent decisions, thereby enhancing the practical application of sensing information in welding.

4. DL-driven techniques for intelligent welding perception and parameter modeling

Intelligent control of welding processes depends on effective sensing, information acquisition, processing, and feature extraction. However,

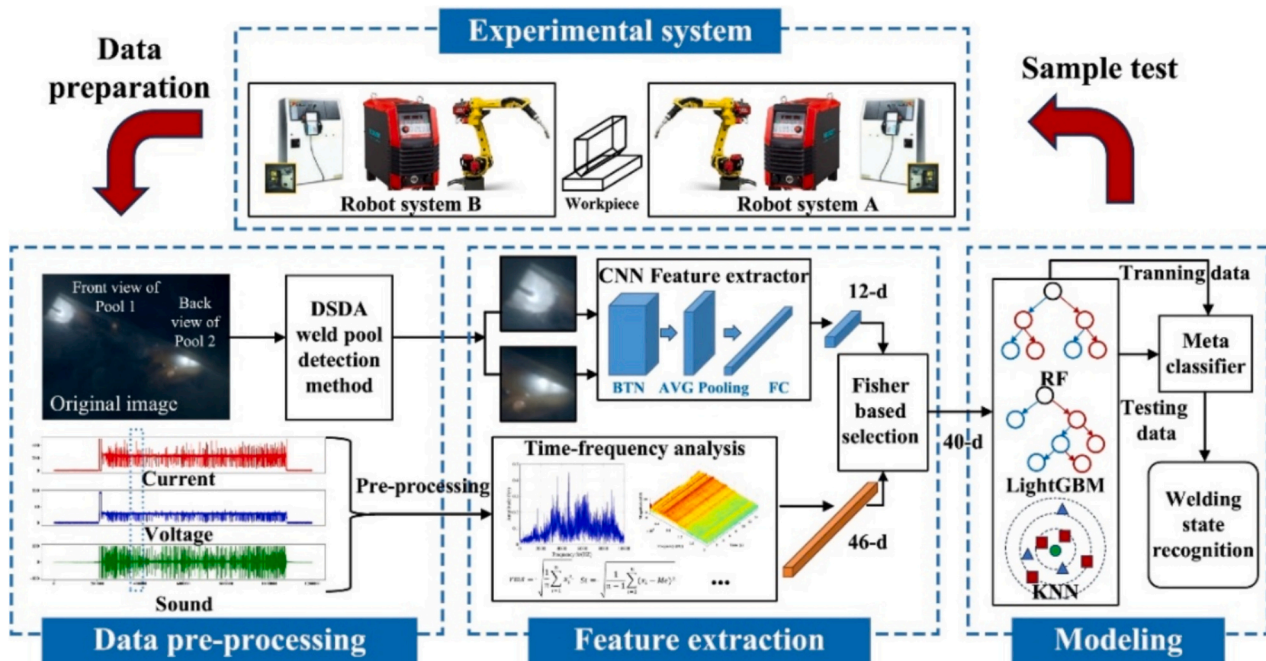


Fig. 16. Overall flowchart of welding status monitoring based on multi-source signals [91].

traditional methods based on sensor design, image processing, and parameter computation demand extensive expertise and involve intricate algorithm development. These constraints limit both the optimization of detection targets and the generalizability of the algorithms. Owing to the highly complex and dynamic nature of the welding process, the sensing information during welding is also diverse and time-varying. Effectively extracting and leveraging this complex information to achieve high-quality welding represents the future direction of intelligent technology. Research areas such as data collection techniques [10,98], data transmission methods [99], data processing technologies [100–103], and data expert systems [11] are pivotal for advancing welding intelligence. Data processing in welding is inherently complex and critical, driving extensive research efforts, especially in DL-based methodologies.

Recently, the advantages of DL—such as superior nonlinear mapping and model induction capabilities—have been extensively utilized for image data processing in welding applications. In pipeline welding, employing CNN models alongside a specialized image processing algorithm enables the real-time, robust extraction of laser line features and

the end-to-end computation of feature parameters. Concurrently, all-position pipeline welding typically requires multi-layer and multi-pass operations. RNNs can learn from and retain information from preceding image data, thereby facilitating the estimation of weld seam feature parameters for subsequent welding layers. This provides a basis for WTPP planning and facilitates intelligent control throughout the multi-layer and multi-pass welding process.

This section outlines the key applications of DL technology in welding and explores its integration with sensing technologies to improve weld quality and enable intelligent control in pipeline external welding. In welding processes, supervised neural network frameworks are generally utilized. Based on the obtained welding data and annotation files, the model learns the mapping relationships for various tasks based on the specified network architecture. The corresponding CNN-based data processing framework is shown in Fig. 17. Research related to image feature processing—such as target classification, detection, segmentation, and tracking—and parameter calculation and prediction within the welding domain is examined.

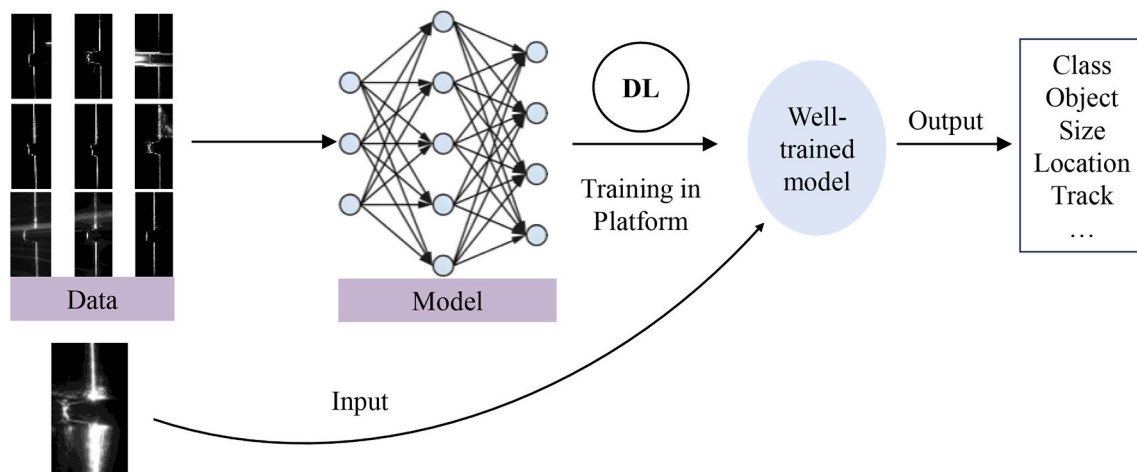


Fig. 17. Data processing paradigm of DL in the welding domain.

4.1. Semantic analysis and feature recognition methods for welding images

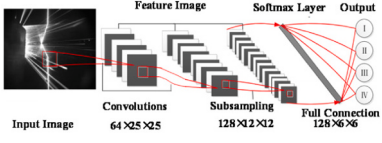
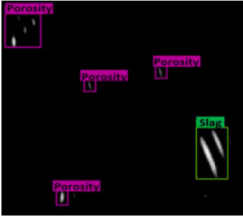
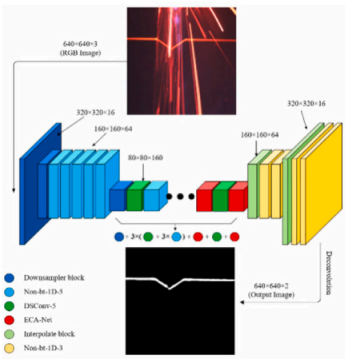
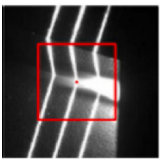
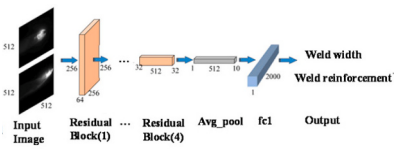
The successful implementation of DL in welding hinges on real-time and stable feature extraction. As detailed in Section 3, vision sensing and associated image processing have been widely investigated and applied during the “Pre-weld,” “In-weld,” and “Post-weld” stages. Consequently, DL technology has become increasingly crucial across various tasks. Numerous neural networks based on CNN have made significant progress in classification, detection, segmentation, object tracking, and related fields. The prevalent networks used for multiple tasks in the welding sector are listed in Table 5.

4.1.1. Target classification

Target classification in welding images involves recognizing welding groove types, assessing penetration states, and classifying weld defects. Predominant network architectures utilized include CNNs, noted for their strong spatial feature extraction; MobileNet, valued for its

lightweight structure suitable for edge deployment; and the ResNet series, where residual connections prevent gradient vanishing and accelerate model convergence. Tian et al. [148] utilized two laser sources alongside a CNN model to categorize groove types, achieving a 97.6 % recognition accuracy. This model was trained via supervised learning using labeled groove data gathered under controlled settings. Du et al. [104] developed a 6-layer CNN to identify groove types amidst substantial interference. Their models were enhanced for generalization through data augmentation and dropout regularization, ensuring more robust classification performance in real-world conditions. During welding operations, DL is commonly employed to determine the penetration state of the molten pool. For instance, Xue et al. [107] applied a MobileNetV2 backbone integrated with a C&R-PSO module to dynamically optimize hyperparameters. This strategy allows for lightweight deployment suitable for real-time applications, particularly in high-speed laser welding. In post-weld inspection, DL-based defect classification is crucial. Miao et al. [115] introduced a two-stage framework that combines edge enhancement preprocessing with a VGG16 classifier,

Table 5
Use of networks in DL for welding domain tasks.

Task types	Type images	Specific tasks	Application Network
Target classification		Welding groove type recognition	CNN [104,105]
		Molten pool melting zone state recognition	CNN [106], Mobilenet [107], RBF [108], ANN [109,110]
		Weld seam defect classification	CNN [111–113], DCNN [114], VGG16 [115,116], CNN-LSTM [117]
Target detection		Laser spot positioning	CNN [118]
		Molten pool and keyhole detection	CNN [119], YOLOv4 [120]
Target segmentation		Welding defects detection	YOLO [121–123], Faster RCNN [124], Mask RCNN [125]
		Laser stripe segmentation	DCNN [126], FCN [127], ERFNet [128], DeepLab [129]
		Molten pool and spatter segmentation	DCNN [130], TNN [131]
		Weld seam area segmentation	U-Net [132], ERFNet [133]
Target tracking		Welding defect segmentation	U-Net [134], Mask RCNN [135], Deeplab [136]
		Weld seam feature point tracking	CNN [37], SSD [38], LSTM [137], RCNN [32], Mask RCNN [138]
Parameter calculation and prediction		Weld seam, molten pool width, and reinforcement height calculation	ResNet [139–142], RNN [119], ANN [143]
		Weld bead, molten pool width parameter, and welding state prediction	ATT-LSTM [144], CNN-LSTM [145,146], GRU [147]

utilizing transfer learning to improve convergence on limited labeled defect data. Zhang et al. [111] crafted a 7-layer custom CNN trained end-to-end on comprehensive image microstructure data, attaining 100 % accuracy due to the uniformity of high-resolution inputs and the controlled conditions of post-weld imaging. To improve target classification in noisy welding environments, attention-based LSTM and transformer models have been deployed to capture temporal patterns. Unlike frame-based approaches, these architectures process sequences of molten pool images or arc signals, thereby enabling superior recognition of dynamic weld states such as underfill, burn-through, and arc instability. This temporal modeling enhances robustness and accuracy, especially when affected by metal spatter, arc light, or motion blur.

4.1.2. Target detection

Detection tasks include identifying weld position, formation status, and weld defects. Commonly used models for these tasks include SSD, the YOLO series, and Faster RCNN. These models strike an optimal balance between the detection accuracy and inference speed, with the YOLO series being particularly favored for real-time industrial inspection owing to its anchor-free design and high throughput. Lu et al. [149] improved SSD by integrating MobileNetV3, enabling real-time detection of manual welding types while reducing computational costs. Lin et al. [118] used a CNN for initial localization of the laser spot, subsequently achieving precise localization through template matching. Gantala et al. [121] and Yang et al. [123] demonstrated the effectiveness of YOLO-based models for weld defect detection. Gantala applied YOLOv4 to identify porosity and slag inclusion across six types of butt joints, whereas Yang employed YOLOv5 to detect eight categories of defects in steel pipe welds. These studies highlight the merits of YOLO architecture in balancing the detection speed and accuracy; thus, this architecture is highly suitable for real-time inspection tasks in complex welding scenarios. Target detection in welding confronts challenges such as arc interference, small defect sizes, and complex backgrounds. Enhancements such as multi-scale feature fusion and high-resolution inputs improve the precise localization of fine targets [150]. Future research may focus on adaptive detection models trained with augmented weld datasets to increase robustness in industrial environments.

4.1.3. Target segmentation

Segmentation tasks include laser stripes, weld seams, molten pools, and defects segmentation. Common models, including the U-Net, DeepLabv3+, and FCN, are preferred for their ability to capture global context and intricate details through encoder-decoder architectures and skip connections. These frameworks are particularly suitable for pixel-level prediction tasks within complex welding settings, where accurate localization is critical. Pixel-level image segmentation allows for the extraction of relevant dimensional measurements, providing a basis for quantitative control. Yang et al. [126], Zhang et al. [127], and Wang et al. [128] addressed laser stripe segmentation amidst significant interference for real-time weld seam tracking by employing lightweight CNN variants trained with synthetic augmentation to ensure both robustness and speed. Tan et al. [131] used a parallel CNN and TNN architecture to achieve real-time metal spatter segmentation, effectively balancing spatial detail with global context. Yang et al. [132] enhanced U-Net with a targeted data augmentation strategy to improve generalization in weld seam segmentation, particularly under varying lighting and metal spatter conditions. For weld defect segmentation, Zhu et al. [134] developed a lightweight segmentation network that integrates feature extraction, attention mechanisms, defect localization, and boundary anti-aliasing modules. Their model, trained on mixed-scale data and optimized with boundary-sensitive loss functions, facilitated the accurate delineation of laser welding defects of varying sizes at high speeds. These segmentation models provide high accuracy and low-latency inference, making them well-suited for real-time control in welding automation systems. Moreover, they have been successfully deployed in practical applications, such as an omnidirectional wall-

climbing robot designed for inspecting spherical tanks. In this scenario, an attention-enhanced Mask RCNN was utilized to enable real-time recognition and robust tracking of circumferential weld seams on curved surfaces [151].

In segmentation tasks, balancing segmentation accuracy, temporal consistency, and rapid inference is a critical challenge. The primary strategies employed to address this include:

1. Designing lightweight network architectures (e.g., MobileNetV2 or EfficientNet-B0) to maintain semantic understanding while reducing computational costs;
2. Enhancing temporal consistency through temporal feature fusion to improve detection stability across image sequences;
3. Implementing attention optimization mechanisms, such as CBAM or Efficient Attention, to focus the model on key weld regions and enhance robustness without compromising real-time performance;
4. Applying model quantization and inference acceleration techniques to increase computational speed.

Experimental validation demonstrates that a lightweight DeepLabv3+ model with a MobileNetV2 backbone and a CBAM attention module, when deployed on an embedded platform, achieved real-time inference at 21 fps and maintained a high average precision of 98.5 % for weld seam identification under noisy, high-reflectivity conditions [90].

4.1.4. Target tracking

The current method primarily involves first detecting the target area, then calculating the target position to perform the tracking task [152]. Research on target tracking within the welding domain focuses on monitoring the weld seam during the welding process. Common network architectures used in tracking models include CNNs for feature extraction, LSTMs or RNNs for temporal sequence modeling, and Mask RCNNs or SSDs for spatial localization. These architectures are favored for their ability to capture both spatial and temporal features, ensuring robust tracking in noisy or dynamic environments. Zou et al. [37] initially employed a CNN to extract welding image features and subsequently introduced a multi-correlation filtering method for real-time seam feature tracking. Later, an LSTM was utilized to model image sequence relationships, allowing the system to maintain tracking continuity despite arc interference or seam occlusion [137]. Furthermore, SSD combined with RNN was used to learn contextual seam features from images, achieving robust tracking under noise interference [38]. Xiao et al. [32] used the RCNN for seam type identification and ROI detection, then achieved stable seam tracking through specifically designed seam center extraction algorithms. Li et al. [138] applied the Mask RCNN to detect seam positions in spherical tank welding and subsequently extracted the weld seam centerline to guide the robot in completing the tracking task. A primary challenge in target tracking for welding is maintaining consistent localization across rapidly changing welding scenes, such as arc interference, spatter, or occlusion. To overcome this, tracking-by-detection frameworks are increasingly being adopted. These systems combine rapid detection frameworks, such as SSDs, with temporal modeling networks such as LSTMs, RNNs, or Transformer models to enhance continuity and robustness across image sequences.

Deploying DL models in complex environments—including heterogeneous materials (e.g., aluminum alloys and stainless steel), noise-prone settings, or subsurface defect detection—requires improving the model's robustness and domain generalization capabilities. This can be achieved by optimizing the model architecture through the following strategies:

1. Domain adaptation and data augmentation: Employ synthetic domain shifts through texture randomization, adversarial domain

adaptation, and the use of extensive cross-material datasets to improve generalization.

2. Robust feature extraction: Replace standard backbones with more resilient alternatives (such as ResNet with attention modules or MobileNetV3) and utilize hybrid CNN-transformer architectures to more effectively learn invariant features.
3. Multi-modal input fusion: Integrate data from various sensor modalities using parallel encoder branches and cross-attention mechanisms to reduce dependence on single-view image signals.
4. Fine-tuning with few-shot samples: Implement few-shot transfer learning using a limited number of target domain samples to adapt models to new environments.

Collectively, these strategies enable DL models to maintain high accuracy and stability in real-world, unstructured welding scenarios [153–155].

In the pipeline welding process, extracting image features—such as laser line characteristics and molten pool boundaries—can also be achieved using the aforementioned network architectures and optimization techniques. Compared with traditional methods, DL-based image feature extraction offers enhanced stability, accuracy, and adaptability, thereby establishing a robust foundation for subsequent process regulation and control. During multi-sensor data fusion, discrepancies can arise between vision-based weld detection and arc signals in areas with significant metal spatter or shadow occlusion. In such cases, strategies such as dynamic confidence evaluation of different sensors and temporal consistency verification effectively preserve output continuity. DL technology also shows considerable promise in addressing these issues. Fusion networks that utilize attention mechanisms or Transformer architectures can dynamically adjust the weighting of each input stream based on signal quality. In addition, some systems integrate rule-based decision logic or reinforcement learning-driven conflict resolution mechanisms at the decision level, significantly enhancing the stability and reliability of the final output. Further improving the capability to recognize complex multiple targets, enhancing multi-source information fusion and decision-making, and fully leveraging spatiotemporal information from welding image sequences alongside other sensor data are crucial for executing high-quality pipeline welding tasks. These capabilities are essential not only for applying DL technology in welding but also for revealing the advancing trends in intelligent welding technology.

4.2. Intelligent modeling and prediction methods for weld formation parameters

Welding parameter calculation and prediction involve understanding the characteristics of the welding process and computing the welding parameter information represented by the current features. The application tasks and corresponding networks are detailed in Table 5.

In the welding domain, feature parameter calculation typically involves learning the extensive features present in images. This is generally combined with the welding target segmentation task, where relevant parameters are computed based on the number of segmented pixels. Wang et al. [139] integrated GC blocks into the ResNet-34 architecture to enhance global feature perception, thereby enabling precise segmentation of weld width and reinforcement height. These features were subsequently mapped to their corresponding actual values. Jiao et al. [106,119] used a CNN alongside a contour extraction algorithm to ascertain backside weld width from molten pool arc images. They employed label-efficient training methods and data augmentation strategies to improve generalization across varied lighting conditions. Wang et al. [142] developed a Res-Seg network for molten pool edge detection and weld width prediction. This architecture features a decoder with skip connections that preserve edge details, achieving a prediction error below 0.2 mm. In addition to image-based features, learning methodologies can be extended to one-dimensional

data. Li et al. [143] developed an ANN to predict weld width and reinforcement height using welding speed, current, and voltage as inputs. To enhance real-time adaptability, the ANN was trained with batch normalization and momentum-based optimization, accelerating convergence and stabilizing online learning.

However, the computation of these welding parameters solely considers spatial information between pixels within a single image, neglecting the sequential (temporal) information across images. RNNs such as LSTM networks can effectively process sequential input data by storing past information along with the current input to determine the current output, thereby achieving precise predictions. Additionally, incorporating an attention mechanism into the model allows for dynamic adjustment of focus on historical temporal information. The typical attention-based time-series prediction model is illustrated in Fig. 18 [144]. Initially, the model utilizes LSTM layers to sequentially extract temporal features, followed by an attention mechanism that focuses on critical segments, such as periods of rapid data change within the sequence. This strategy enhances the sensitivity of the model toward essential temporal dependencies, thereby improving prediction accuracy and stability.

In welding parameter prediction, neural networks model nonlinear relationships and subsequently compute the necessary feature parameters based on the newly obtained inputs. Chen et al. [145] used a CNN to extract visual features from molten pool images and combined them with arc voltage, welding current, and arc power into a 19-dimensional feature vector. An LSTM network was then used to learn the time-series characteristics of these fused features, enabling the model to capture both spatial and temporal correlations across heterogeneous sensing modalities. This architecture ensures consistent predictions even amidst arc-induced fluctuations or visual occlusions. They achieved a prediction accuracy of up to 99.25 % across various welding states, including normal fusion, incomplete fusion, indentation, burn-through, and misalignment. The specific network architecture is shown in Fig. 19. However, standard LSTM assigns equal attention to all time steps, potentially diluting critical transient patterns. To address this, a temporal attention mechanism was introduced to dynamically focus on time points more strongly correlated with defect formation. Ye et al. [144] applied an Attention-based LSTM (ATT-LSTM) model that integrates keyhole, arc, and molten pool features to predict the back weld width, achieving a prediction error within 0.5853 mm. This model significantly enhanced robustness and prediction stability in complex welding environments.

RNNs effectively leverage temporal sequence information between images and, compared with CNNs, provide more accurate calculations of output parameters. In multi-layer and multi-pass pipeline welding, designing an appropriate network structure is crucial as it facilitates the effective utilization of image data sequences and enables accurate, stable computation of key parameters. Multi-layer multi-pass welding in pipelines involves continuous welding across layers, where the welding torch's motion on the welding robot exhibits strong repetition and similarity between successive layers. Fig. 20 illustrates the torch trajectory relative to the weld seam center during multi-layer welding using an orbital welding robot. The welding torch trajectory remains largely unchanged owing to the robot's repetitive motion along the track during successive layer welding. Trajectory variations between different layers are primarily caused by groove shrinkage resulting from thermal deformation during welding. When the torch operates across consecutive layers, it can learn from the trajectory data of preceding layers to guide the welding of subsequent layers.

This can be achieved through the following strategies:

1. Utilizing attention mechanisms: When forecasting current welding parameters, RNN/LSTM models enhanced with attention mechanisms can automatically focus on previously welded regions that exert greater influence, including zones with local path deviations or thermal distortions. This mechanism increases the ability of the

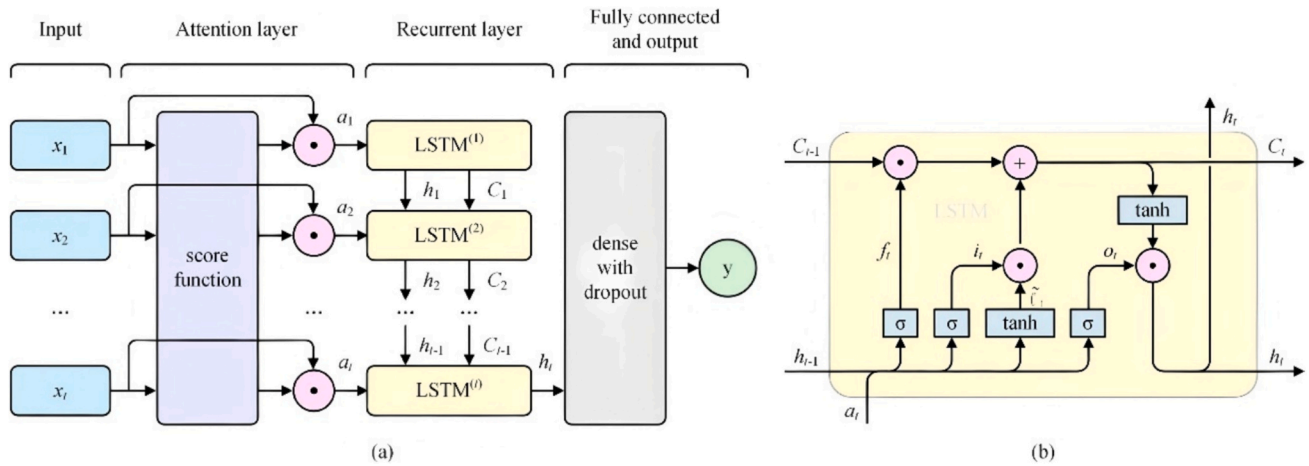


Fig. 18. Schematic of ATT-LSTM neural network structure [144].

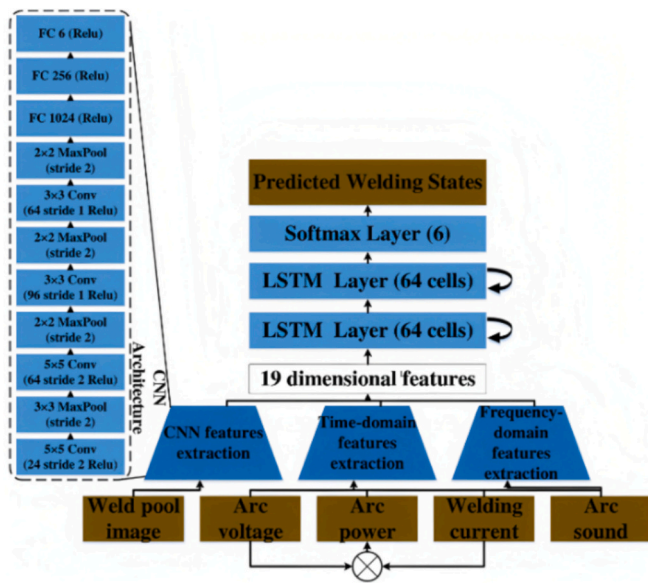


Fig. 19. CNN-LSTM network structure [145].

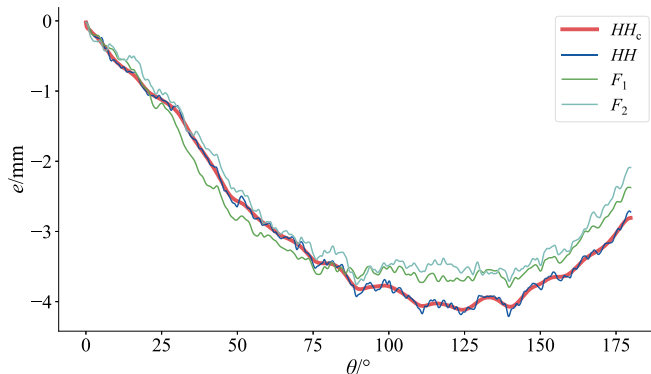


Fig. 20. Schematic of pipeline multi-layer (first three layers) welding path and welding-torch control curve.

model to capture essential historical information, thereby improving overall prediction accuracy and stability [156].

- Utilizing Transformer-based architectures for layer-by-layer online learning: Advances in attention mechanisms have enabled

Transformer-based architectures (such as Transformer and Informer; typical structures are depicted in Fig. 21 [157]) to outperform traditional LSTM + Attention models in multi-pass welding parameter prediction. Transformers use self-attention to capture long-range dependencies across input sequences without relying on recurrence, allowing for parallel training and inference. This results in faster training times and greater efficiency over extended time frames. Informer, an optimized variant, employs sparse attention and probabilistic sampling techniques to reduce computational complexity in long-sequence forecasting. These architectures support online layer-by-layer learning during multi-layer welding, allowing the model to discern global patterns in real time and dynamically update after each layer, thereby preventing error accumulation in sequential pass predictions [158,159].

- Uncertainty quantification: Methods such as Monte Carlo Dropout, can be used to evaluate prediction confidence in layer-by-layer parameter prediction for multi-layer welding, thereby reducing error propagation [160].
- Feedforward-feedback hybrid framework: In multi-layer welding processes, incorporating prior insights from DL-based predictive models with MPC can optimize the welding trajectory in advance. This approach enables the pre-computation of control actions, promoting smoother and more intelligent torch movements. It also helps prevent abrupt adjustments, thereby facilitating anticipatory feedforward planning. Concurrently, sensor feedback is integrated via an enhanced Kalman filtering method to achieve optimal estimation of the current system motion state, allowing for stable feedback correction [161]. The resultant optimized welding torch motion trajectory is represented by the HH_c curve in Fig. 20. This curve displays that integrating feedforward and feedback strategies within the hybrid framework effectively reduces prediction and control errors, leading to more stable and accurate torch motion control. These strategies provide improved support for WTPP planning and welding parameter adjustments, thereby further enhancing the intelligence of pipeline welding processes.

Nevertheless, acquiring high-quality annotated data for training intelligent welding models remains a significant challenge. The demanding welding environment—characterized by high temperatures, intense arc light, metal spatter, and reflections—often produces noisy or distorted images, making the annotation of weld features such as molten pool boundaries and defects a time-consuming, costly task that requires specialized expertise and substantial manual effort. Moreover, the diverse range of welding scenarios necessitates large and varied datasets to ensure model generalization. Collecting such data consistently and safely in real production settings is difficult. To overcome these

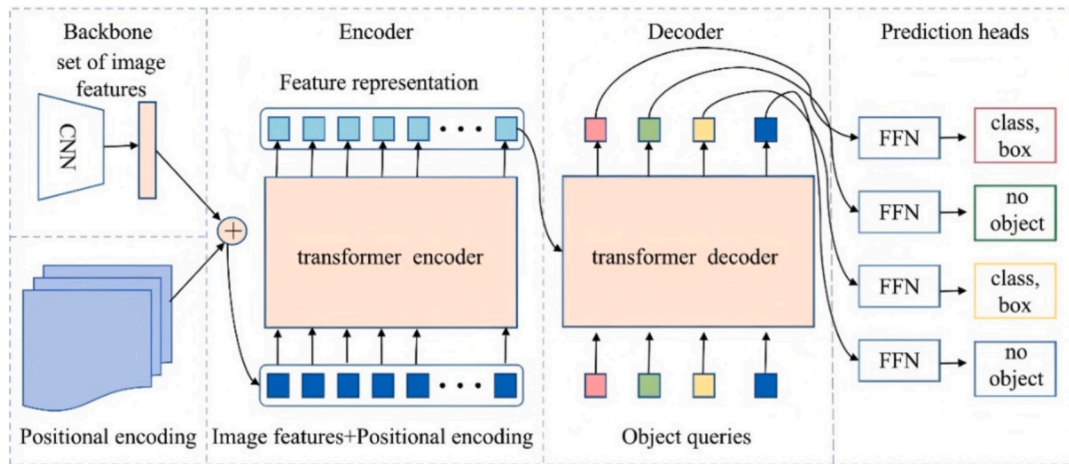


Fig. 21. Architecture of detection Transformer neural networks in welding defect detection [157].

challenges, researchers are investigating: 1. Semi-supervised learning, which reduces dependence on manual annotations by primarily training on mostly unlabeled data combined with limited labeled samples [162]; 2. Data augmentation involves applying transformations and noise to increase data variability and enhance robustness; 3. Simulation-based synthetic data generation entails creating large, diverse, and accurately labeled virtual datasets tailored for hazardous or hard-to-access welding conditions [163]. These methods improve model generalization while reducing data collection costs, thereby accelerating the deployment of intelligent welding systems.

As a foundational technology in artificial intelligence, DL has become a pivotal force in advancing intelligent solutions for complex welding tasks and has been extensively examined, as depicted in Fig. 22. Similarly, other intelligent technologies—such as digital twins, data augmentation, big data, and human-machine interaction—are increasingly being applied in the welding domain. When multiple technologies are used simultaneously, appropriate evaluation mechanisms must be considered to prevent data overfitting. In addition, practical strategies should be devised to ensure the effectiveness and reliability of the model in on-site deployment systems. Comprehensively addressing these aspects can substantially enhance welding intelligence, paving the way for stable and reliable unmanned operations.

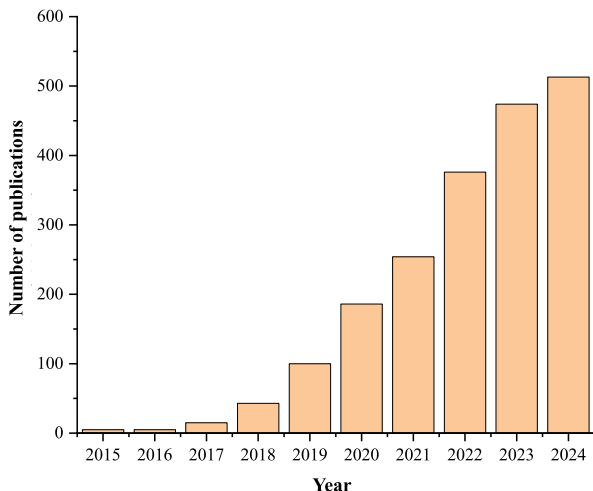


Fig. 22. Worldwide literature (search terms: welding & deep learning).

5. Summary and prospect

In this paper, the current development status of welding technology and equipment used in pipeline on-site welding, the widespread application of vision sensing, and the significance of multi-sensor fusion are summarized and analyzed. The application and potential of DL technology in advancing welding intelligence are also elaborated. The future development of intelligent pipeline welding is shown in Fig. 23 and can be approached from the following aspects:

- (1) The advancement of intelligent welding technology and equipment for pipelines. To enhance the efficiency and quality of pipeline external welding, existing welding technologies and equipment must be developed and upgraded. For example, in welding large-diameter pipes, increasing the number of welding torches or wires is feasible as long as the heat input remains well controlled. Additionally, research should focus on overcoming challenges associated with applying advanced technologies to on-site pipeline welding, such as laser-arc hybrid welding, to achieve technological innovation in pipeline welding methods. Integrating multi-sensor fusion technology with specially designed pipeline welding robots that feature comprehensive WTPP control, an intelligent welding system can be established to facilitate real-time monitoring and regulation of the welding process. These technological advances are expected to significantly improve the efficiency and quality of pipeline external welding in the future.
- (2) The integration and development of multi-sensor technology combined with vision sensing. Vision sensing, known for its extensive detection capabilities, has been widely researched and implemented. Its success depends on meticulously designed sensor architectures and robust image processing algorithms suitable for complex welding environments. Combining vision with other sensing modalities enables more comprehensive perception, while multi-source data fusion enhances the accuracy and reliability of decision-making. This fusion facilitates intelligent control of WTPP and welding parameters, ultimately achieving high-quality welds formation. Advancing multi-sensor fusion remains a key research priority for future intelligent welding systems.
- (3) Research and application of intelligent technologies based on DL. Applying intelligent technologies, particularly DL, is essential for advancing welding automation. Developing an efficient and lightweight CNN architecture enables real-time recognition and robust processing within complex welding environments. In addition, integrating temporal models such as RNNs and

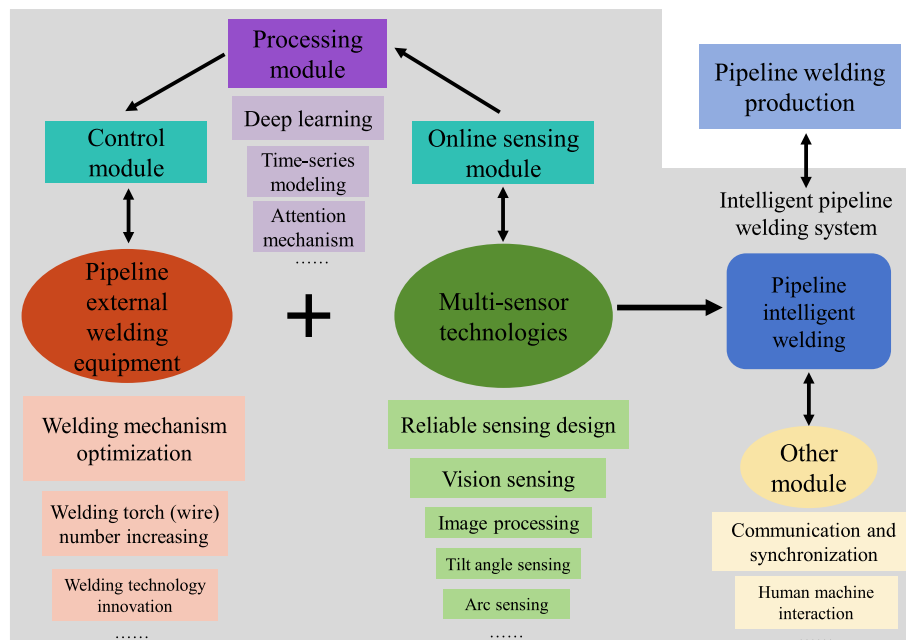


Fig. 23. Future development direction of pipeline intelligent welding.

Transformers, along with attention mechanisms, allows the system to learn and prioritize critical and historical information from multi-source data sequences. This facilitates accurate assessment of welding conditions and reliable evaluation of ongoing welding tasks. Furthermore, integrating emerging technologies—including digital twins, big data analytics, data augmentation, and human-machine interaction—tailored to specific task requirements, can further enhance the intelligence of welding systems. These advancements collectively support the realization of fully intelligent and automated welding processes.

CRediT authorship contribution statement

Chuanhui Zhu: Writing – review & editing, Writing – original draft, Software, Methodology, Investigation. **Zihao Wang:** Writing – review & editing, Methodology. **Zhiming Zhu:** Supervision, Funding acquisition. **Qingzhi Zheng:** Writing – review & editing, Methodology. **Jichang Guo:** Writing – review & editing.

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Declaration of competing interest

The authors declare that they have no known competing financial interests or personal relationships that could have appeared to influence the work reported in this paper.

Data availability

Data will be made available on request.

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